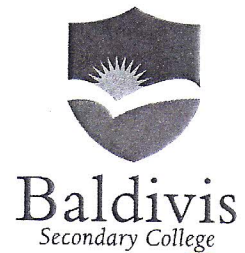


Year 10
Pathway 2



Test 1

Name: Marking Key

Total Marks: 24 marks
Total Time: 25 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Calculator and note sheet (1 A4 Page) allowed

1.

[5]

The planet Pluto is 5.9×10^9 km from the Sun. The distance from Earth to the Moon is 3.84×10^5 km.

(a) Write both distances as expanded numbers.

Pluto: 5900000000

Moon: 384000

(b) How many Earth–Moon distances can fit between the Sun and Pluto. Write the answer in standard form, correct to two decimal places.

$$5900000000 \div 384000 = 15364.58333 \\ = 1.54 \times 10^4 \quad (2dp).$$

(c) Assume that the Moon's orbit is a circle. What is the circumference of that circle? Give your answer in scientific notation, correct to three decimal places.

Formula $C = 2\pi r$

$$C = 2\pi 384000 = 2412743.158 \\ = 2.413 \times 10^6 \quad (3dp)$$

2. How much will \$8000 be worth if it has been invested for 3 years with an interest rate of 2% p.a. compounding yearly? Give your answer correct to the nearest dollar.

[2]

$$A = P \left(1 + \frac{r}{100n} \right)^{nt}$$

$$A = 8000 \left(1 + \frac{2}{100} \right)^3 = \$8490$$

3. A \$10 000 investment is placed in a bank returning 6% compound interest. Find the value of the investment, correct to the nearest cent, after a year if interest is calculated every:

(a) six months

[2]

$$A = P \left(1 + \frac{r}{100n} \right)^{nt}$$

$$A = 10000 \left(1 + \frac{6}{100 \times 2} \right)^2$$

$$= \$10,609$$

(b) month.

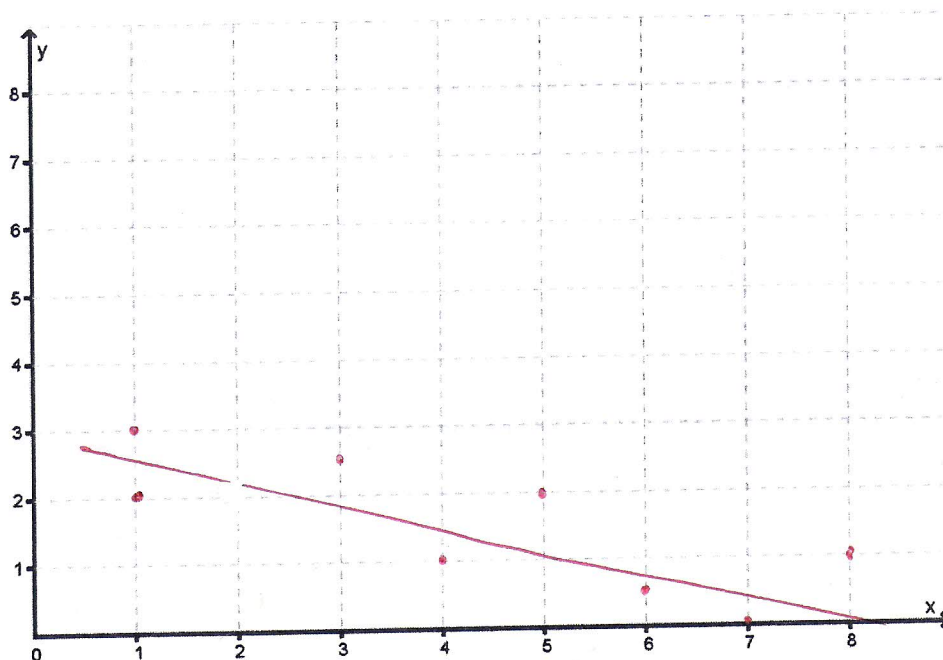
[2]

$$A = 10000 \left(1 + \frac{6}{100 \times 12} \right)^{12}$$

$$= \$10,616.78$$

4. Plot the following points onto the graph, eye in the line of best fit and describe the relationship between x and y. [5]

(1, 3), (1, 2), (3, 2.5), (4, 1), (5, 2), (6, 0.5), (7, 0), (8, 1)

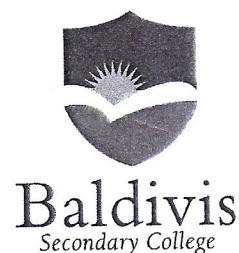


linear
strong
negative
relationship

or

as x increases
y decreases

Year 10
Pathway 2



Test 1

Name: Marking key.

Total Marks: 34 marks
Total Time: 30 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Calculator free

1. (a) Simplify leaving your answer in positive index form $(a^{-4}b^3)^3$ [2]

$$a^{-12}b^9 = \frac{b^9}{a^{12}}$$

- (b) Evaluate 5^{-2}

$$\frac{1}{5^2} = \frac{1}{25}$$

2. Write each of the following in simplest index form:

a. $(9^2)^7$

$$9^{14}$$

b. $x^2 \times 2x \times 3xy$

$$6x^4y$$

c. $\frac{12a^4}{3ab}$

$$\frac{4}{b}$$

d. $(7 \times 5)^5$

$$7^5 \times 5^5$$

or 35^5

3. Write a set of data with at least 10 values that has a lowest value of 2, a highest value of 30, lower quartile of 12, upper quartile of 20 and median of 18. [5]

A range of answers are possible:

2, 3, 4, 12, 14, 15, 18, 19, 19, 20, 22, 23, 30

* must have minimum 10 values or $\frac{1}{2}$ mark if only shows 5 values.

4. Use words from the list below to complete the following sentences.

[5]

~~interest~~ ~~principal~~ ~~compound interest~~ ~~appreciation~~ ~~depreciation~~

(a) Simple interest depends on the principal, interest rate per annum and the time in years.

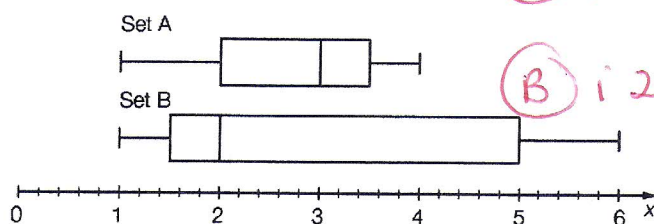
(b) The original amount of money loaned or invested is called the principal.

(c) Interest that is calculated on the principal and interest from the previous time period is called compound interest.

(f) Increase in cost or value is called appreciation and decrease in cost or value is called depreciation.

5. For each data set find (i) median (ii) range and (iii) interquartile range.

[7]



(A) i 3

ii $4 - 1 = 3$

iii $3.5 - 2 = 1.5$

(B) i 2

ii $6 - 1 = 5$

iii $5 - 1.5 = 3.5$

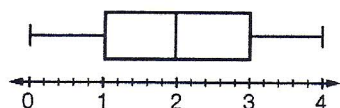
(b) Compare the statistics for the two data sets.

Set B median is lower.
Set B has a positive skew.

Any other reasonable response.

6. The five-number summary for this box plot is:

[1]



A 1, 2, 3, 4, 5

B 0, 1, 2, 3, 4

C 0, 1, 1, 2, 4

D 0, 1, 2, 2, 4

5. a) Find the mean, median and IQR of the scores for Maths and English tests.

[6]

	28	30	32	39	40	41	42	50
English	40	42	28	30	50	39	41	32
Maths	42	48	24	32	45	14	39	40

Maths:

✓ mean = $\frac{284}{8} = 35.5$

✓ median = 39.5

✓ IQR = 15.5

English:

✓ mean = $\frac{302}{8} = 37.75$

✓ median = 39.5

✓ IQR = 10.5

(b) Comment on the statistical values calculated and compare English and Maths results. [2]

Median is the same.

✓ Maths mean is lower than the English mean.

IQR for maths is higher.

Any other reasonable response.

* At least 2 points needed.